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Introduction and Dataset Setup – Brian

This lab has allowed me to learn about overfitting, underfitting, and bias-variance trade-offs. This lab aimed to understand how a variety of different machine learning models work using Python libraries such as NumPy, Matplotlib, and Scikit-learn. I have gained an understanding of how to create synthetic data and models. This lab has allowed me to gain an understanding of how different models will produce different results depending on how complicated they are. I have also learned the importance of choosing the right model that will provide accurate predictions.

I was responsible for importing the required libraries and creating synthetic data in this first code cell. I have added comments to help explain my code. While creating synthetic data may seem like an easy part of this lab, I have learnt that it is good to get everything set up correctly, as the rest of the code depends on this. It has given me an understanding of what the first steps of a machine learning project are like. In future projects, I would like to improve my knowledge of how to create and prepare datasets.

Polynomial Regression – Toan

For the polynomial regression section of the lab, I worked with three different polynomial degrees: 1, 4, and 15. I tested each model and compared the graphs to see how well they fit the dataset. Each degree produced a different regression curve, which allowed me to observe how

changing the polynomial degree affected the model's performance. From this comparison, I learned that a more complex model does not always provide better results.

After reviewing the graphs, I noticed that the degree 1 model was too simple to capture the overall pattern of the data, which caused underfitting. On the other hand, the degree 15 model was too complex and followed the data points too closely, resulting in overfitting. The degree 4 model provided a better balance because it followed the general pattern of the dataset without being too simple or too complex.

Before completing this activity, I did not fully understand the differences between underfitting and overfitting or how they affected a model's performance. However, testing the models and comparing their graphs helped me understand these concepts more clearly. I also realized how important it is to choose an appropriate polynomial degree instead of simply using a higher one. In future projects, I can apply what I learned to evaluate different models and select one that best represents the data.

Learning Curves – Montserrat

I worked on the learning curves section, which helped me compare the training scores and cross-validation scores of different models. These graphs showed me how the models performed as the amount of training data increased. I used the graphs to understand how well the models learned from the training data. This gave me another way to identify the differences

between overfitting and underfitting. It also helped me understand why checking a model's performance is important.

From this section, I learned that an underfit model usually performs poorly on both training and validation data. An overfit model performs much better on training data than on validation data. I found that learning curves are useful because they help show whether a model can perform well on data it has not seen before. This connected to what I learned in class about training and testing machine learning models. In the future, I can use learning curves to evaluate models and see where improvements are needed.

Group Reflection and Collaboration - Brian

I collaborated with my group members when answering the knowledge check questions. We discussed the difference between overfitting and underfitting. I used the knowledge I gained from looking at the graphs and model scores to help answer these. This meant I could explain my understanding of the topics whilst also hearing other people's ideas. This meant I could look back at the information from all parts of the lab. Collaborating with my group members helped me understand what the purpose of the assignment was.

I think I could have improved my communication when completing each section. This would have helped me understand the whole lab rather than just my section. I learnt how going through other people's work can help me spot any errors they may have made and understand their results better. Splitting up the work made the assignment easier to complete, but improving my communication when discussing each section would have improved my teamwork skills. I gained a better understanding of machine learning by sharing ideas and comparing results.

Overall, I have learnt the importance of teamwork and communication when completing a technical project.

Discussion of Improvements and Learning - Toan

Throughout this lab, I gained a better understanding of how machine learning models work and how different levels of complexity can affect their performance. I learned how to use polynomial regression to compare models and how learning curves can help identify whether a model is performing well. I also became more familiar with Python libraries such as NumPy, Matplotlib, and Scikit-learn, which helped me analyze data and create graphs to visualize the results. Overall, this lab gave me a chance to apply what I learned in class to actual coding activities and better understand the concepts through practical examples.

For future machine learning projects, I can use what I learned from this lab to test different models and evaluate their results before deciding which one works best. I also learned that checking both training and validation scores is important because a model that performs well on training data may not perform as well on new data. Another thing I could improve is being more involved in reviewing my group members' work and discussing their findings. This would help me understand the entire project instead of focusing only on my assigned section. I believe the knowledge and experience I gained from this lab will be helpful as I continue studying artificial intelligence and machine learning.

Conclusion - Montserrat

Lab 08 helped me understand the importance of finding the right balance when building machine learning models. By comparing polynomial regression models and learning curves, I was able to see how underfitting and overfitting affect model performance. I learned that a model that is too simple may not recognize important patterns in the data. I also learned that a model that is too complex may not perform well on new data. These activities helped me understand why choosing the right model is important.

Overall, this lab improved my knowledge of machine learning and gave me more experience working with Python and data visualization. I learned that a good model should be able to recognize patterns without simply memorizing its training data. I can use these concepts in future assignments and machine learning projects. I also learned the importance of communication and dividing responsibilities when working as a group. This experience gave me a better understanding of how machine learning models are evaluated and improved.